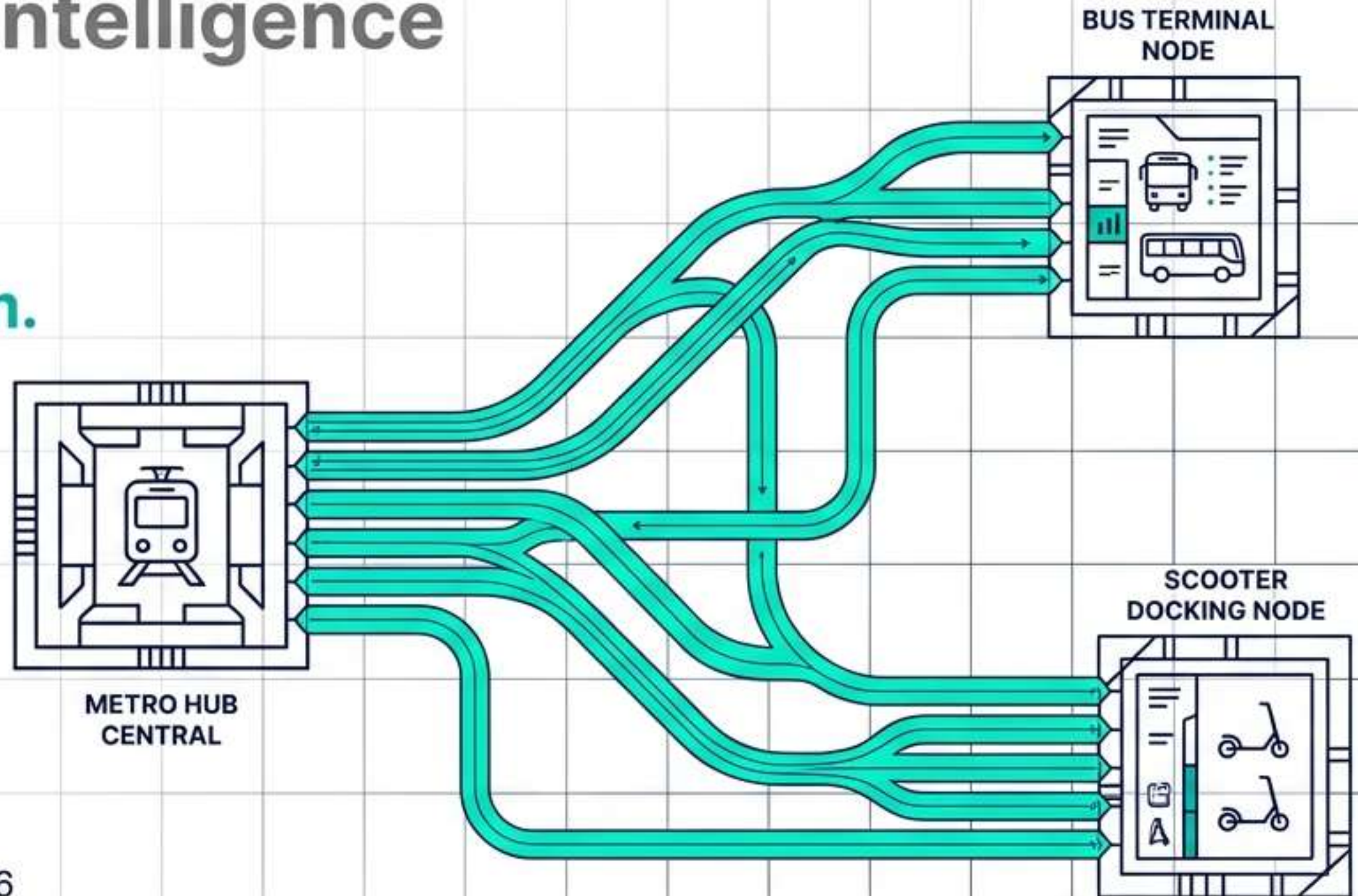


OneJourney AI

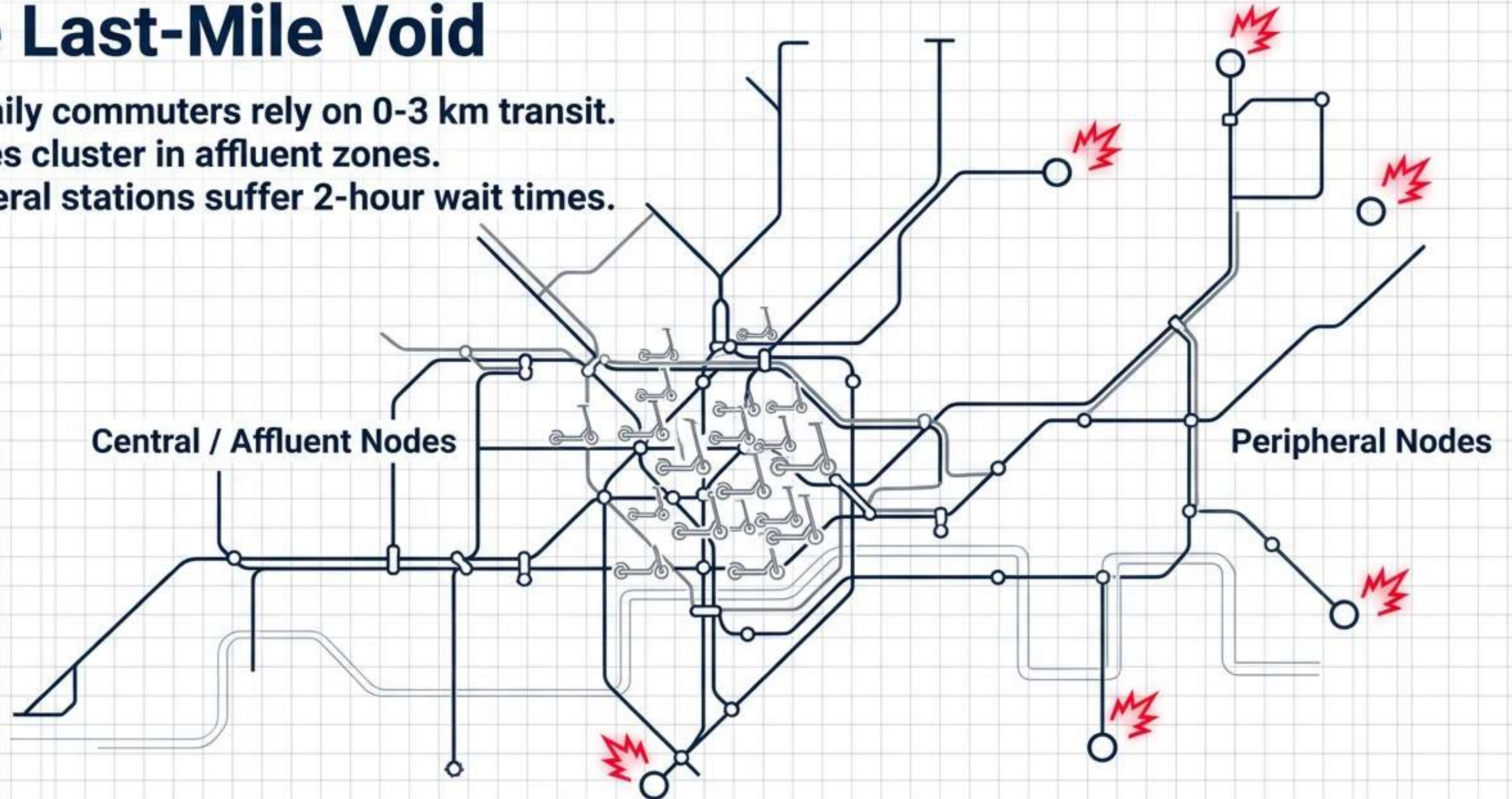
Predictive Last-Mile Intelligence for Urban Mobility

Predict shortages before
commuters experience them.



The Last-Mile Void

47M daily commuters rely on 0-3 km transit.
Vehicles cluster in affluent zones.
Peripheral stations suffer 2-hour wait times.



Wait Time: 2 Minutes

Wait Time: 120 Minutes

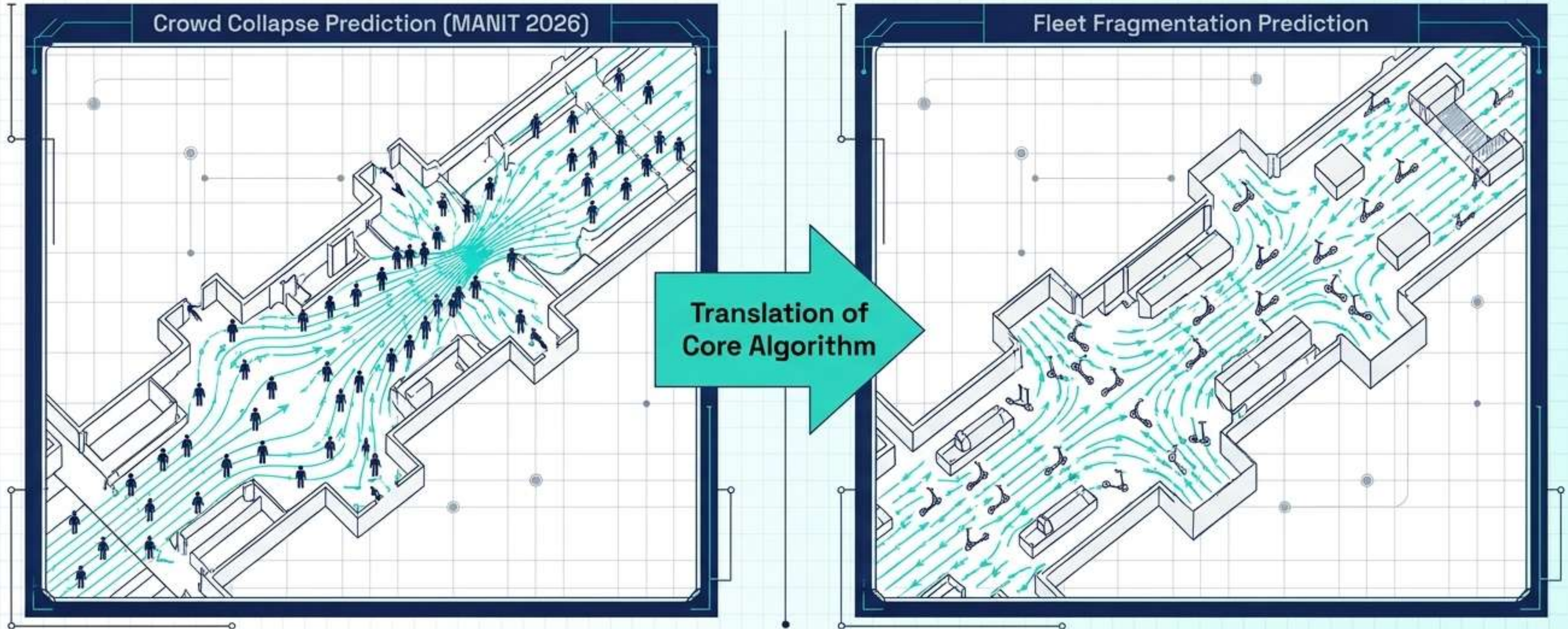
Diagnostic Matrix: Reactive vs. Predictive

Current routing optimizes individual trips. Network resilience requires preemptive coordination.

	Current Reactive Systems	OneJourney Predictive Architecture
Action Trigger	Reactive (After Shortage)	Predictive (15-min Before)
Architecture	Centralized $O(n^3)$	Decentralized $O(n)$
Equity	Ignored	Mathematical Constraint

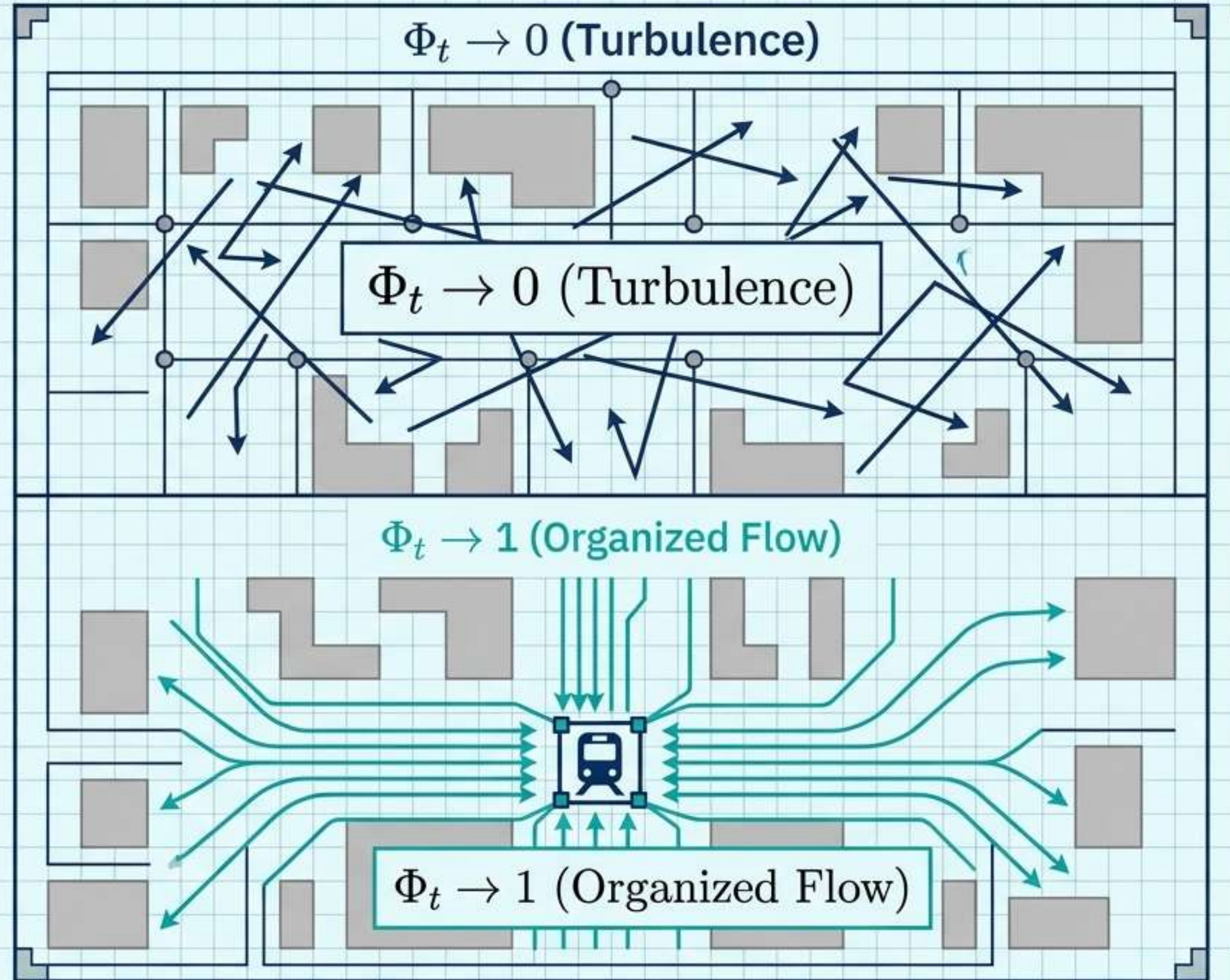
Bio-Inspired Predictive Intelligence

We adapt validated crowd-collapse dynamics to micromobility. Measuring fleet coherence predicts availability collapse before physical shortages manifest.



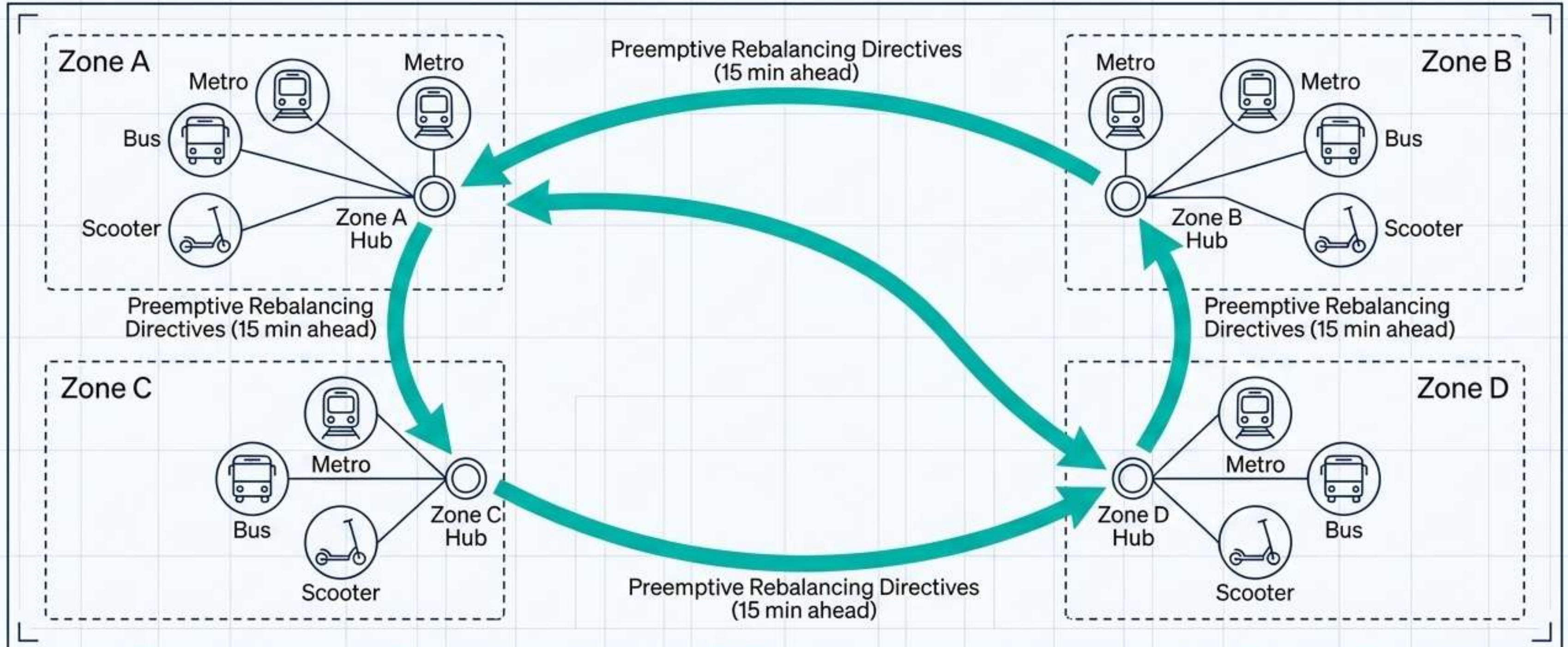
The Demand Flow Field

Request trajectories transform into origin-destination vectors. High coherence indicates healthy flow; low coherence signals impending fleet collapse.



Decentralized Multi-Agent Architecture

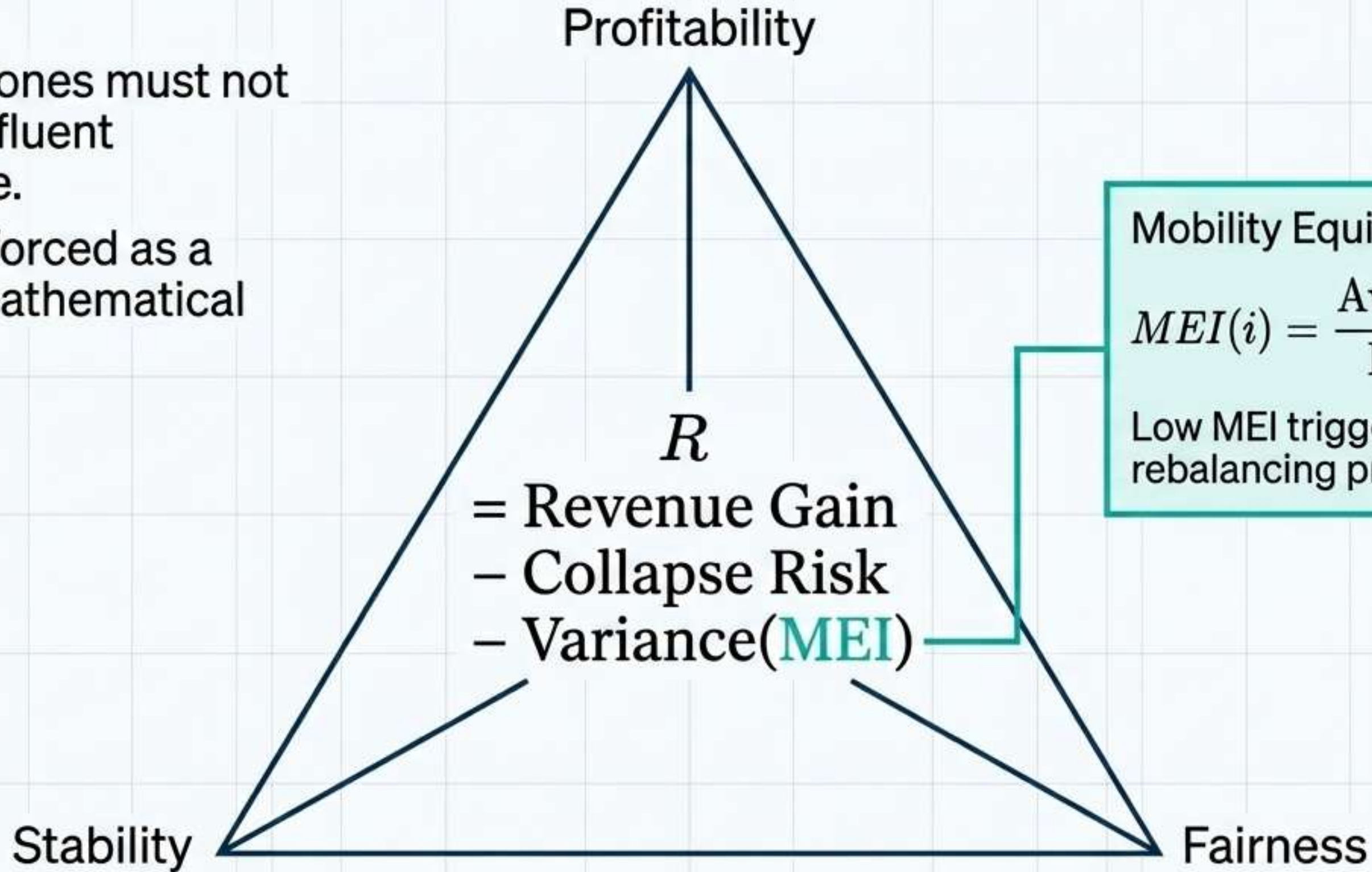
Independent zone agents replace centralized controllers. **Preemptive communication** enables $O(n)$ scalability across transport modes.



The Mobility Equity Engine

Peripheral zones must not subsidize affluent convenience.

Equity is enforced as a structural mathematical constraint.



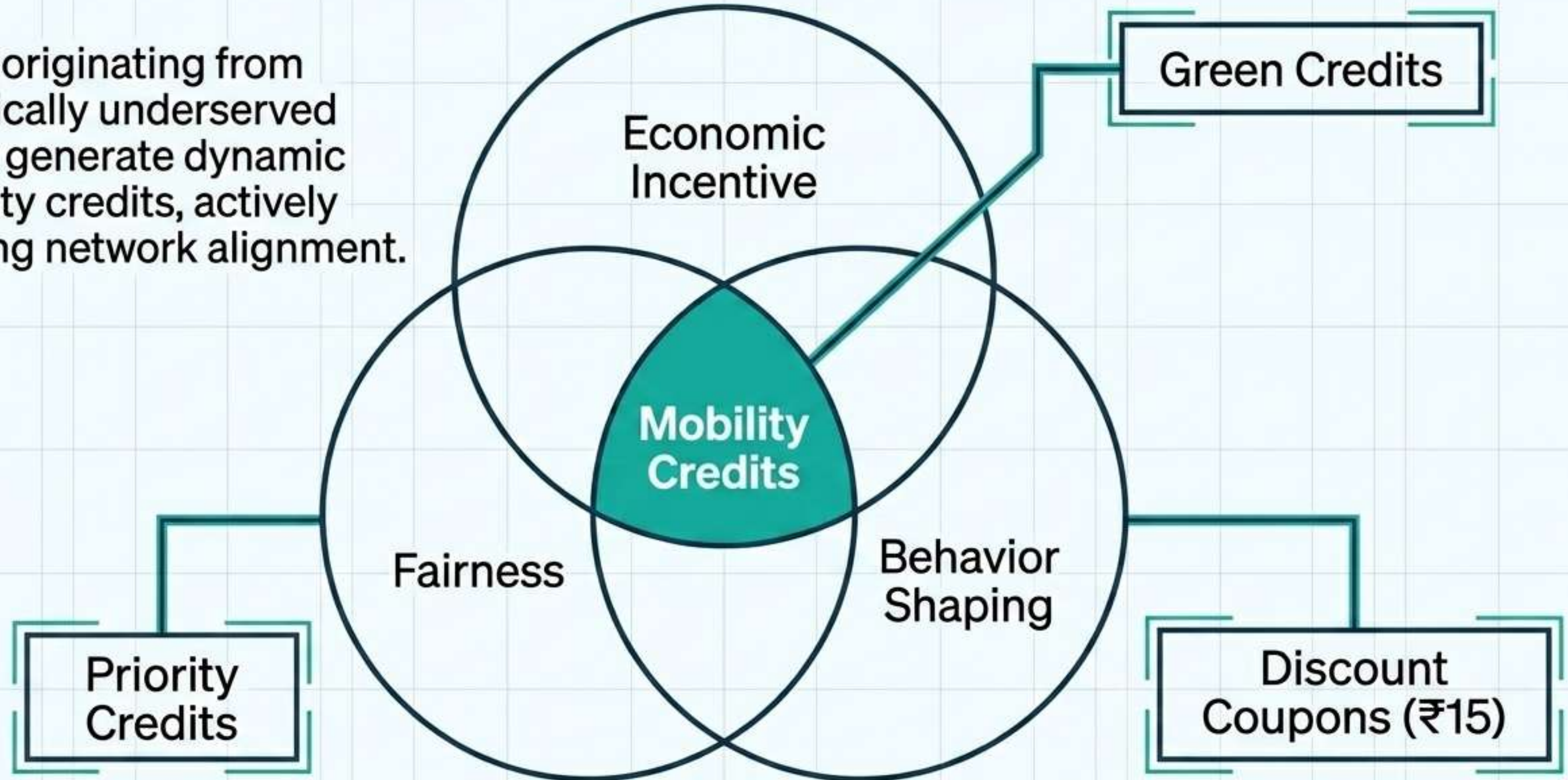
Mobility Equity Index:

$$MEI(i) = \frac{\text{Availability}(i)}{\text{Demand}(i)}$$

Low MEI triggers automatic rebalancing prioritization.

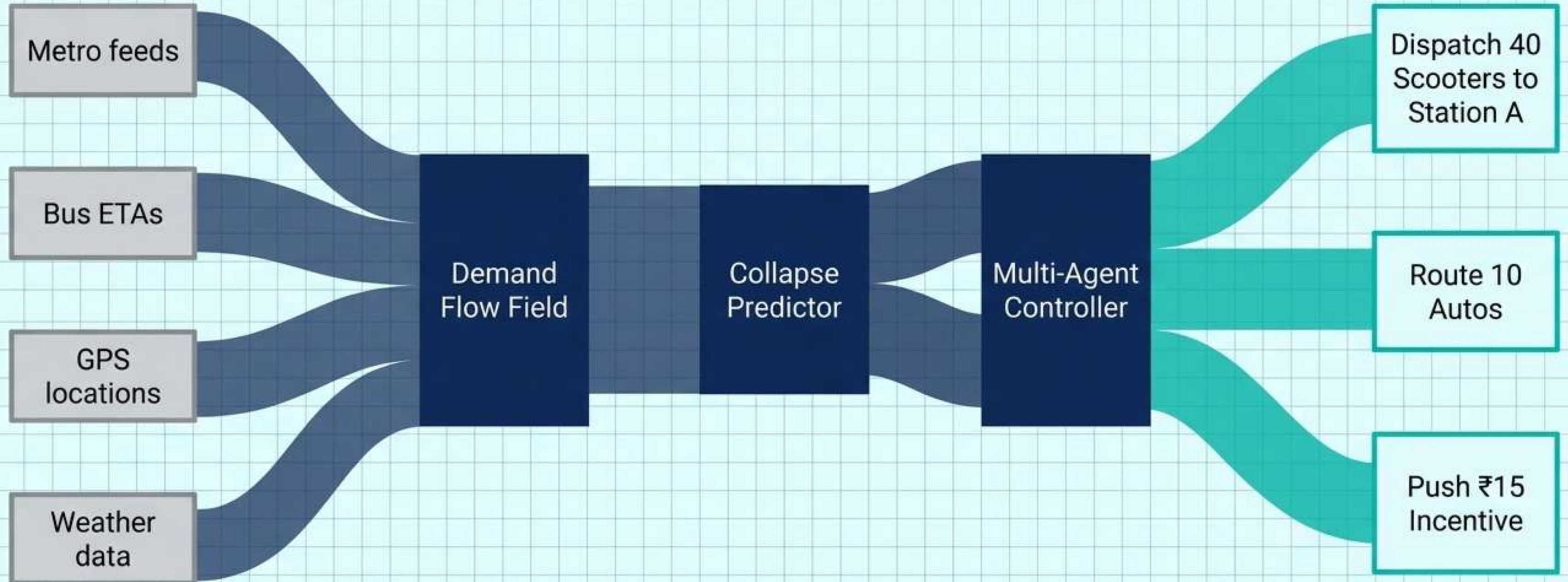
Resolving the Digital Credit Paradox

Users originating from historically underserved zones generate dynamic mobility credits, actively shaping network alignment.



End-to-End Predictive Workflow

From fragmented transit data to proactive physical interventions.



Impact, Assumptions, and Future Work

A B2B infrastructure layer optimizing fleet utilization and equity.
Real-world mobility validation remains future academic work.

Impact & Business



- **Target:** ₹5-10L/mo per city SaaS licensing.



- **Metric:** < 15% wait time variance.

Limitations

Laboratory Disclaimer:

Simulation Mode Active.
Algorithm validated via historical
API stubs. Closed-loop field
testing pending.

From reactive commuting to predictive mobility.

Assumptions and Limitations

Consolidating weaknesses and theoretical constructs for research maturity.

Theoretical Frameworks & Hypotheses

- **Cross-Domain Transfer:** Framework inspired by crowd-collapse research; not experimentally validated for urban mobility.
- **Demand Flow Field:** Conceptual hypothesis; coherence Φ_t is an experimental indicator requiring empirical validation.
- **Prediction Horizon:** Optimal horizon undefined; currently a short-term window (5–30 mins).
- **Complexity Claims:** Asymptotic claims ($O(n^3)$ vs $O(n)$) not mathematically established; decentralized architecture improves scalability.
- **Conceptual Reward Function:** Weighting coefficients remain undefined; further optimization needed.

Data & Implementation

- **No Real-World Data:** Prototype operates in Simulation Mode using synthetic datasets.
- **MARL Simplification:** Agents assumed to cooperate; ignores competing incentives, noisy estimates, partial observability.
- **Digital Twin Limitation:** Performance under real urban conditions is unknown.
- **Explainability:** Interpretability mechanisms for policy recommendations are not yet incorporated.
- **Generalization Unproven:** Parameters learned in one city may not transfer due to differing traffic patterns.

Scope & External Factors

- **Equity Index Calibration:** MEI is a first-order fairness metric; ignores income, demographics, and temporal patterns.
- **Algorithmic Bias:** Demand-driven optimization may amplify inequality without additional constraints.
- **Preliminary Concepts:** Mobility credits are future extensions; operational costs not explicitly minimized.
- **External Shocks:** Robustness under rare events (weather, protests, accidents) is unexplored.
- **Business Assumptions:** Revenue estimates are illustrative; no pilot studies or market validation conducted.

Overall Scientific Position: Current stage is **Research Prototype (TRL 2–3)**. Demonstrated conceptual architecture and simulation workflow. **Future Work** includes real-world validation, data integration, explainability, cost optimization, and pilot deployment.
No claim of state-of-the-art performance, optimality, or production readiness.